CODESHIELD 2026 — A CYBER ODYSSEY

Layered Presence Authentication Framework (LPAF)

Team Name: Cyber Sentinels
Team Members: Sourjak Das & Shivani Tangudu
Domain: Cybersecurity

Core Mission

Presence is no longer claimed — it is **cryptographically verified.**

Problem Statement

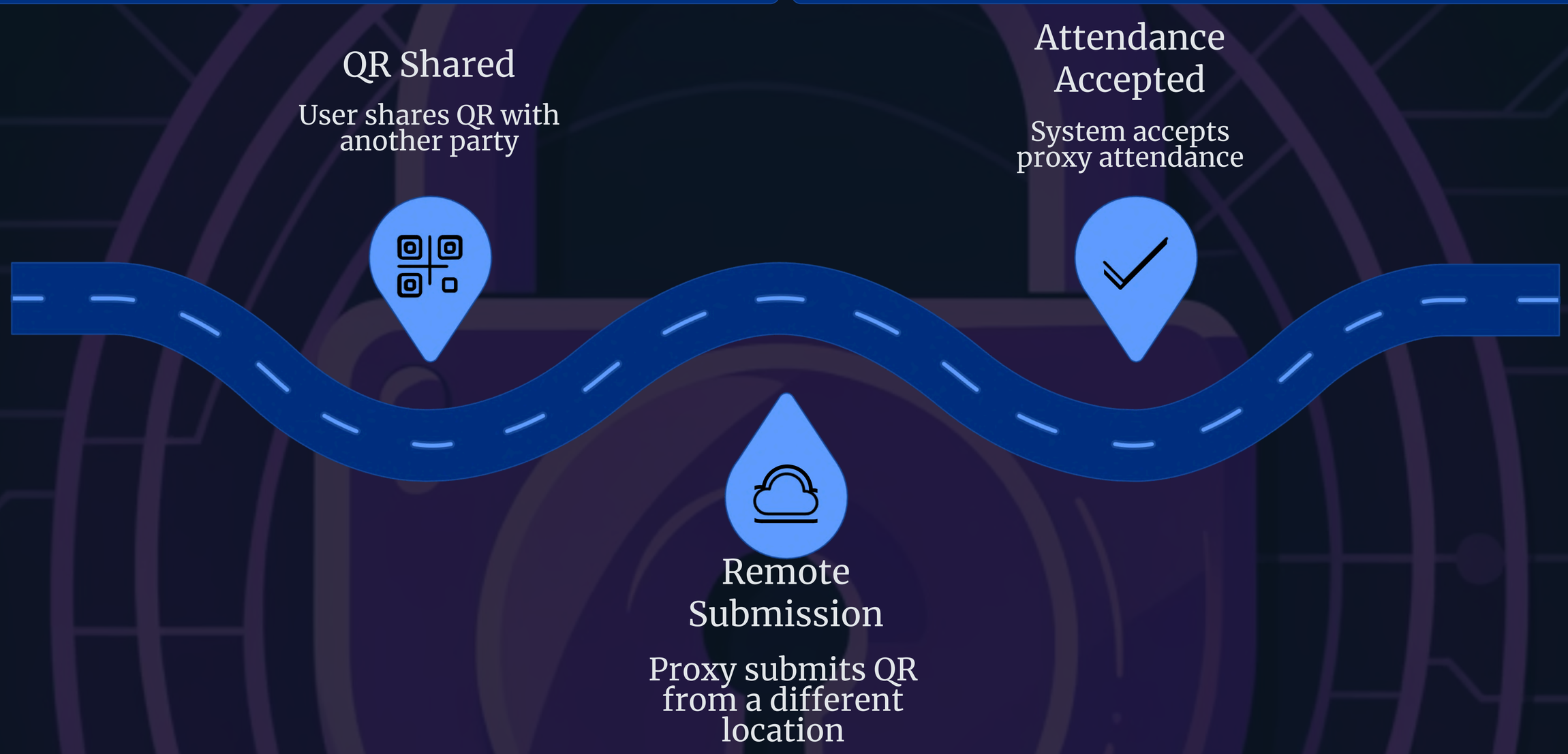
Current attendance systems rely on static QR codes, shared links, and simple online forms. These allow proxy attendance — where students can mark attendance without being physically present.

Unauthorized Marking
Attendance marked remotely with no physical presence check

Academic Data Manipulation
Records can be falsified with no audit trail

No Physical Verification
Zero confirmation that the student is on campus

Weak Authentication
Shared links and static QR codes offer no real security



This simple chain illustrates how today's systems are trivially exploited — a shared QR or link is all it takes to fraudulently mark attendance from anywhere in the world.

Proposed Solution

LPAF applies **cybersecurity principles to attendance verification**, transforming a simple form submission into a **cryptographically verified presence event.**

1

Dynamic QR Token

Time-expiring, session-bound tokens that cannot be reused or shared

2

Network Validation

Campus Wi-Fi IP check confirms physical presence on-site

3

Session Authentication

Session-bound tokens prevent cross-session replay attacks

4

Duplicate Prevention

Roll number deduplication blocks multiple submissions per student

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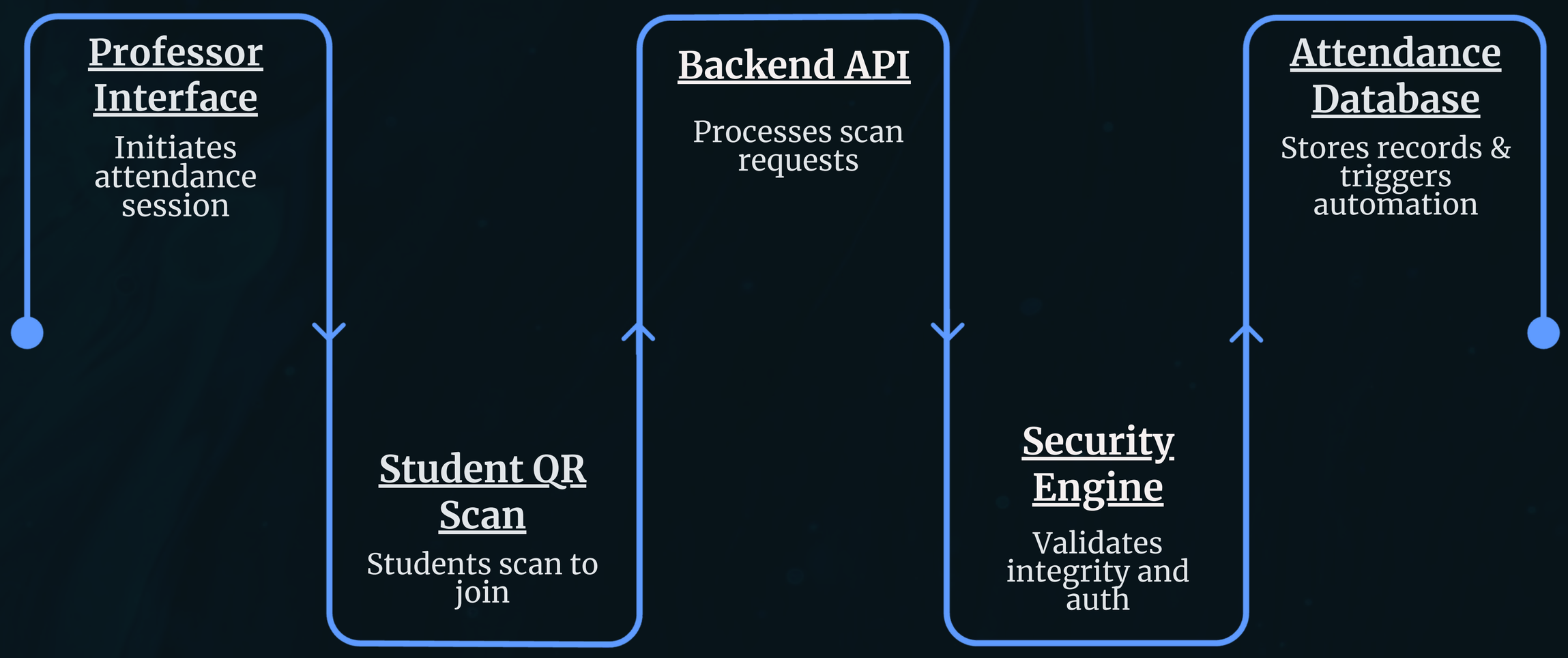
System Architecture & Security Workflow

LPAF is built on a **5-layer vertical architecture**, each layer independently enforcing security while contributing to a seamless end-to-end attendance workflow. This demonstrates **technical depth and implementation feasibility** using proven cybersecurity principles.



Each layer is independently responsible for a distinct function, ensuring that a failure or bypass at one layer does not compromise the integrity of the entire system.

1

Layer 1 — Professor Interface

Professor opens the demo page and enters session information: **Professor email, Subject code, Year, Section, Department.** The system generates a **unique session ID and dynamic QR code** tied exclusively to that session.

2

Layer 2 — Student Interface (QR Scan)

Students scan the QR code using their mobile devices. A mobile-friendly page opens where the student enters their **Name** and **Roll number**. The student submits attendance **without installing any app**.

3

Layer 3 — Backend API Layer

Technology: **Python Flask API.** Responsibilities: Generate session tokens, encode QR data, receive student submissions, and trigger the validation pipeline.

4

Layer 4 — Security Validation Engine

Each submission passes through multiple checks: **Token validation, Time expiration validation, Campus Wi-Fi IP verification, Duplicate roll number detection.** Only when *all* checks pass is attendance recorded. This follows **defense-in-depth security principles**.

5

Layer 5 — Database & Automation Engine

Validated submissions are stored with: **Student name, Roll number, Submission timestamp, Session information.** Once the session ends, the **automation pipeline begins**.

Security Validation Deep Dive

Defense-in-Depth Model

No single check is sufficient. LPAF stacks four independent validations so that bypassing one layer does not grant access. Every submission must pass **all four gates** before being recorded.

Validation Gates

- **Token Validation** — Is the QR token legitimate and session-bound?
- **Time Expiration** — Has the token window expired?
- **Wi-Fi IP Check** — Is the device on the campus network?
- **Duplicate Detection** — Has this roll number already submitted?

Automation & Impact

Beyond secure verification, LPAF introduces **end-to-end attendance automation** — a complete automated academic workflow that eliminates manual attendance compilation entirely.

Automation Workflow

Start Session

Generate QR

Students Submit

Store & Report

Once the session ends — either manually or via timer — the automation pipeline triggers immediately, requiring zero additional action from the professor.

Retrieve All Submissions

Backend collects all validated student submissions from the session database

Compare Against Roll Numbers

System cross-references submissions against the expected roll number list for the section

Identify Absentees

Missing roll numbers are automatically flagged and marked absent

Generate Excel Report

A structured, formatted Excel attendance report is generated automatically

Email to Professor

The report is sent directly to the professor's registered email — no manual steps required

Benefits, Use Cases & Future Scope

Benefits

- Prevents proxy attendance
- Ensures real physical presence verification
- Automates attendance reporting
- Creates audit-ready attendance records
- Reduces manual work for professors

Use Cases

- Universities and colleges
- Examination halls
- Corporate training programs
- Professional certification centers
- Secure events

Future Scope

- AI-based anomaly detection
- Device fingerprint verification
- Real-time attendance analytics
- ERP integration for institutions

5
Security Layers
Independent validation gates every submission must pass

0
Manual Steps
Required from professor after session ends — fully automated

4
Validation Checks
Token, time, Wi-Fi IP, and duplicate detection per submission

1
Automated Report
Structured Excel report delivered directly to professor's inbox

Secure verification of presence should be the standard for modern institutions.

> "Attendance is no longer claimed. It is cryptographically verified."

— **Cyber Sentinels** · Sourjak Das & Shivani Tangudu · CodeShield 2026